A. O. M. Shamsuddoha

Lecturer and Program Coordinator, Department of Electrical and Electronic Engineering
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Profile

University lecturer in Electrical and Electronic Engineering with nearly six years of teaching experience and seven peer-reviewed publications, serving as Program Coordinator of the B.Sc. in EEE program at ULAB since 2025. Hands-on expertise in Cadence Virtuoso, Verilog and PSpice. Seeking a PhD in Electrical and Computer Engineering.

Education

M.Sc. in Electrical and Electronic Engineering Coursework completed
Ahsanullah University of Science and Technology (AUST), Dhaka, Bangladesh
CGPA: 3.50 / 4.00

B.Sc. in Electrical and Electronic Engineering December 2018
Ahsanullah University of Science and Technology (AUST), Dhaka, Bangladesh
CGPA: 3.75 / 4.00
Thesis: *Low Power VLSI Circuit (SRAM) Design Using FinFET*. Supervisor: Prof. Satyendra Nath Biswas, PhD

Publications

Journal Articles

[J1] M. Sakib, G. Hossain, T. Mahmud, A. Al Hasan, M. Nafis, M. Y. Hossain, N. M. Chisty, M. M. Mahmud, **A. O. M. Shamsuddoha** and M. M. Hossain, "SleepingEdge: Real-Time Sleep-Stage-Driven Epileptic Seizure Detection Using HMM and Hybrid OptiNet-SVM," *IEEE Access*, vol. 14, pp. 57646–57673, 2026. doi: 10.1109/ACCESS.2026.3682946

Conference Papers

[C1] S. H. M. Atul, R. A. Shium, **A. O. M. Shamsuddoha**, M. Islam, M. A. Hossain and M. Hasan, "An Efficient and Interpretable Hybrid FCM-ELM Framework with Entropy-Guided Feature Selection for Multiclass Brain Tumor MRI Segmentation and Classification," in *2026 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)*, Dhaka, Bangladesh, 2026. Accepted and presented.

[C2] M. M. L. Chowdhury, S. Sarder, M. R. Hossain, M. R. A. Panto, N. A. Mim and **A. O. M. Shamsuddoha**, "A Low-Cost IoT-Based Smart Home Automation and Security System with Energy-Efficient Operation for Sustainable Living in Developing Regions," in *2026 2nd International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS)*, Dhaka, Bangladesh, 2026. Accepted and presented.

[C3] S. H. M. Atul, **A. O. M. Shamsuddoha**, S. M. M. Islam, M. R. Biswas, H. K. F. Hai and S. M. Nawshed, "Detection and Classification of Poultry Diseases Using Deep Convolutional Neural Networks: A Ten-Class Image-Based Approach," in *2025 IEEE 4th International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON)*, Dhaka, Bangladesh, 2025, pp. 462–467. doi: 10.1109/RAAICON69033.2025.11502034

[C4] S. Chakraborty, S. Zahir, N. T. Orchi, M. F. Bin Hafiz, **A. O. M. Shamsuddoha** and S. M. Dipto, "Violence Detection: A Multi-Model Approach Towards Automated Video Surveillance and Public Safety," in *2024 International Conference on Advances in Computing, Communication, Electri-*

cal, and Smart Systems (iCACCESS), Dhaka, Bangladesh, 2024, pp. 1–6. doi: 10.1109/iCACCESS61735.2024.10499466

[C5] S. J. Hamim, M. K. Islam, **A. O. M. Shamsuddoha**, S. Dhar and A. K. M. Baki, "Performance Evaluation of 2×2 and Massive 8×8 MIMO at 28 GHz and 60 GHz for Dhaka City," in *2022 IEEE 7th International Conference for Convergence in Technology (I2CT)*, 2022, pp. 1–5. doi: 10.1109/I2CT54291.2022.9825246

[C6] **A. O. M. Shamsuddoha**, M. K. Islam and S. N. Biswas, "Design and Performance Analysis of SRAM Circuit Using Adiabatic Logic with FinFET," in *2021 International Conference on Electronics, Communications and Information Technology (ICECIT)*, Bangladesh, 2021, pp. 1–4. doi: 10.1109/ICECIT54077.2021.9641167

Academic Experience

Program Coordinator, B.Sc. in EEE Program, Department of Electrical and Electronic Engineering February 2025 – Present
University of Liberal Arts Bangladesh (ULAB), Dhaka

- Plan semester-wise course offerings and class routines, and assign teaching assistants
- Form capstone project groups and organize moderation of examination questions
- Prepare the department's ISO certification and SDG reports
- Coordinate departmental academic meetings, including continuous quality improvement (CQI) sessions, and advise students on academic matters

Lecturer, Department of Electrical and Electronic Engineering June 2023 – Present
University of Liberal Arts Bangladesh (ULAB), Dhaka
Courses: Electrical Circuits I; Electronic Circuits I; Electromagnetic Fields and Waves; Solid State Devices; Power Electronics; VLSI I; Microprocessor and Embedded Systems

Lecturer, Department of Electrical and Electronic Engineering April 2022 – May 2023
University of Scholars, Dhaka
Courses: VLSI I; Microprocessor and Interfacing; Electrical Properties of Materials; Electronic Devices and Circuits; Measurements and Instrumentation
Supervised several undergraduate projects in these areas.

Part-Time Lecturer, Department of Electrical and Electronic Engineering December 2021 – May 2022
Ahsanullah University of Science and Technology (AUST), Dhaka
Courses: VLSI I Lab; Microprocessor, Interfacing and System Design Lab; Energy Conversion II Lab

Lecturer, Department of Computer Science and Engineering May 2019 – April 2020
Daffodil International University, Dhaka
Courses: Electronic Devices and Circuits; Basic Electrical Circuits

Academic Service and Leadership

Technical Reviewer

- International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS), Dhaka October 2026
- International Conference on Computer and Information Technology (ICCIT), Cox's Bazar December 2025
- International Conference on Computer and Information Technology (ICCIT), Cox's Bazar December 2024

- International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS), Dhaka March 2024

Workshops Conducted

- Resource person, "Digital VLSI Circuit Design Essentials: Unleashing Creativity with Cadence Virtuoso Tool," Department of EEE, ULAB, with ULAB Electronics and Robotics Club and IEEE RAS ULAB Student Branch Chapter March 2024

Conference Presentations

- Poster presentation, International Conference on Electronics and Informatics 2022, Bangladesh Electronics and Informatics Society, representing University of Scholars January 2023

Organizing and Advising

- Advisor, ULAB Electronics and Robotics Club, University of Liberal Arts Bangladesh May 2024 – July 2025
- Organizer, International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS), Dhaka March 2024
- Advisor, Robotics Club, University of Scholars November 2022 – May 2023
- Assistant Coordinator, MATLAB & Image Processing Competition, Engenius 2018 (inter-university tech competition), AUST February 2018

Professional Development

- Induction Training for Tertiary Teachers (24 hours, four modules), Center for Excellence in Teaching and Learning (CETL), ULAB Fall 2025
- Two-Day Training on Outcome-Based Education (OBE), BAETE May 2025
- Certificate Course on Teaching for Active Learning, Foundation for Learning Teaching and Research September 2023
- Short course on Planning and Design of Electrical Substation, Continuing Education Center, University of Scholars January 2023
- Training on Outcome-Based Curriculum and Accreditation Standards and Criteria, The International University of Scholars April 2022
- Machine Learning Master Class, Pantech Prolabs India Pvt Ltd April 2021
- Virtual BCI & Neurotechnology Spring School, g.tec medical engineering GmbH, Austria April 2021
- Online Workshop on Quantum Systems Engineering for Scientists, IEEE Quantum February 2021
- CCNAv7 (Cisco Certified Network Associate, version 7), American International University-Bangladesh March – November 2020
- Workshop on Writing Research Papers Using LaTeX, IEEE AUST Student Branch January 2018
- Workshop on Introduction to Basic TCAD, United International University August 2017
- Training on AutoCAD 2D and 3D, CADD CORE Training Institute, Dhaka 2016
- Training on Web Programming Using PHP/MySQL, Institute of Information Technology, University of Dhaka May – July 2016
- Online courses, LinkedIn Learning: Learning FPGA Development; Raspberry Pi: GPIO; AutoCAD Electrical Essential Training; Learning Java; Programming Foundations: Object-Oriented Design; Learning SQL Programming 2020 – 2021

Honors and Awards

- 1st Runner-up, ROBORACE (Line Following Robot), Engenius 2018 inter-university tech competition, AUST EEE Society 2018
- 1st Runner-up, Tech Idea Race 2016 (innovative idea writing competition), IEEE AUST Student Branch 2016
- Perfect Attendance Award, Notre Dame College, for over 98% class attendance 2014

Technical Skills and Languages

VLSI design	Cadence Virtuoso, Microwind
Circuit simulation	OrCAD PSpice, Proteus, Quartus II
Programming and HDL	Verilog, MATLAB, C++, PHP, HTML
Design tools	AutoCAD, Tinkercad
Documentation	Microsoft Word, Excel, PowerPoint
Languages	Bengali (native), English (IELTS Academic 8.0, 2024), Hindi (basic)

Projects and Competitions

- Performance evaluation of 2×2 and massive 8×8 MIMO at two frequencies for Dhaka City March 2021
- Microcontroller-based DC to AC converter July 2018
- Digital door lock system in Proteus February 2018
- 4-bit up/down synchronous counter in Cadence January 2018
- Participant, IEEEXtreme 11.0 Programming Competition October 2017
- Smoke detection system with protection June 2016
- Home automation with GPS April 2016
- Participant, 11th and 12th Bangladesh Mathematical Olympiad (2013, 2014) and 2nd Bangladesh Physics Olympiad (2013)

Also took part in programming contests, innovative idea competitions, engineering olympiads and workshops.

Professional Memberships

- IEEE December 2016 – December 2025
- IEEE Computer Society, Student Member December 2023 – December 2024
- IEEE Power and Energy Society, Student Member December 2017 – December 2018
- IEEE Women in Engineering (WIE), Student Member December 2016 – December 2018
- IEEE AUST Student Branch, Student Member November 2016 – December 2018
- AUST Innovation and Design Club, General Member May 2017 – December 2018

References

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